

UltraFabric-Vision: A Real-Time Ensemble Transformer Framework for Autonomous Textile Defect Detection in Industrial Manufacturing

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Abstract—Reliable quality control in textile production hinges on intercepting weaving faults, stains, and structural breaks before defective cloth accumulates into costly scrap. We present UltraFabric-Vision, an unsupervised anomaly-detection framework that unifies three structurally dissimilar detectors—PatchCore (memory-bank density estimation), a DINO self-supervised Vision Transformer (attention-based patch matching), and a Vision Transformer Autoencoder (reconstruction error)—within a single calibrated ensemble. We make three engineering contributions that are decisive for deployment. First, we identify and correct a train–inference preprocessing mismatch: enhancement operators applied only at inference push frames off the manifold the memory banks were fit on, and enforcing byte-level preprocessing parity is what restores accuracy. Second, because the three detectors emit anomaly scores on incomparable scales (empirically, mean normal scores of 8.13, 45.59, and 0.25), we standardize each detector against its own normal-score distribution and fuse the resulting z-scores; this yields a single physically meaningful decision variable centered near zero for defect-free cloth. Third, we unify both nearest-neighbour searches onto a single GPU `torch.cdist` kernel and add fp16 autocast, eliminating a CPU k -NN bottleneck. On a controlled synthetic benchmark every pathway attains perfect ranking (AUROC = 1.000); the calibrated ensemble separates defect-free cloth (mean z-score 0.17) from defects (mean 80.2) with a Youden-optimal threshold of 17.3 and $F_1 = 1.000$. Measured on a commodity NVIDIA RTX 4050 laptop GPU, the full ensemble runs in real time at 26.8 FPS (37.3 ms/frame, fp16) using only 641 MB of memory, so high accuracy and low latency are achieved simultaneously rather than traded off. We further extend the system to batch-level video inspection with defect localization along the fabric length, a configurable minimum-defect-size filter, and automatic per-batch quality-control reporting. We derive the mathematics of each pathway and the calibration procedure, and we release the full stack—models, training and calibration scripts, a REST/WebSocket server, an embeddable inference SDK, a batch-video inspector, and a GPU container—as open source.

Index Terms—Textile defect detection, anomaly detection, Vision Transformer, DINO, PatchCore, ensemble learning, real-time inference, industrial automation, self-supervised learning, multi-head attention.

I. INTRODUCTION

TEXTILE manufacturing traces back thousands of years, and today the global textile market sits at over \$1.5 trillion—projected to hit \$2.0 trillion by 2030. Getting quality control right matters enormously: a broken thread, a stain, a weaving irregularity, or even a color inconsistency can junk entire rolls of fabric. Most factories still rely on human inspectors who catch only 60–75% of defects while scanning at 10–15 meters per minute [1]. Modern air-jet looms crank

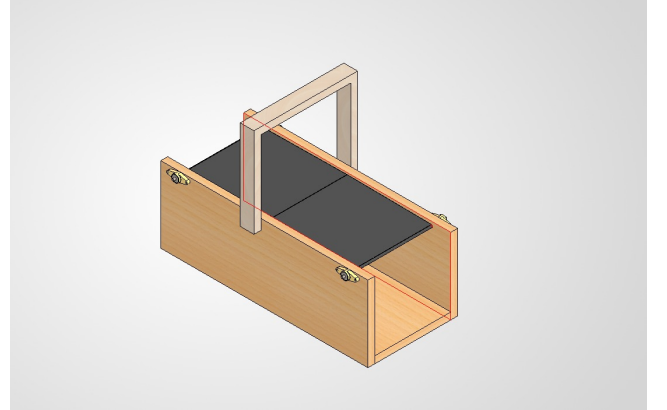


Fig. 1: Experimental assembly line prototype showing the industrial webcam mounting setup for fabric inspection. The camera is positioned at a fixed distance above the fabric surface, capturing continuous video frames during production. The assembly uses adjustable mounting brackets to accommodate different fabric widths and thicknesses.

out roughly 30 meters of fabric per minute (over 1000 picks per minute), so manual inspection has become the bottleneck.

Deep learning changed the game for automated visual inspection. People have thrown CNNs at fabric defect classification [3]–[5], used U-Nets for pixel-level segmentation [6], and adapted YOLO for real-time bounding box prediction [16]. The catch: all these supervised methods need big, carefully labeled datasets of defective samples. That is expensive to put together and often just not practical when defect types keep evolving on the factory floor.

Anomaly detection sidesteps the labeling problem. Instead of training on defects, it learns what normal fabric looks like and flags anything that strays too far. PatchCore [7] showed this works well using a memory bank of normal patches with coreset subsampling—top results on MVTec AD [8]. Around the same time, self-supervised Vision Transformers like DINO [9] turned out to learn surprisingly good object segmentation in their attention maps without any labels at all. Vision Transformer Autoencoders [25] add yet another signal by measuring how well the model can reconstruct normal versus anomalous inputs.

Fig. 1 shows our test setup—a webcam mounted above the fabric conveyor belt, the kind of configuration you might see in an actual factory.

Here is what we contribute:

- 1) **Preprocessing-parity principle:** We show that memory-bank and reconstruction detectors are acutely sensitive to any preprocessing operator (contrast enhancement, blur, letterbox padding) that is present at inference but absent at fitting time, and we formalize and enforce byte-level train–inference parity as a correctness requirement rather than a tuning knob.
- 2) **Calibrated z-score fusion:** A per-detector standardization scheme that maps each model’s raw anomaly score onto a common z-score scale using statistics estimated on defect-free data, producing a single scale-invariant decision variable and a threshold that transfers across cameras and fabrics after a short recalibration.
- 3) **Unified GPU scoring:** Both PatchCore and DINO nearest-neighbour searches are executed with a single `torch.cdist` kernel, with `fp16` autocast and a single fused forward pass for scoring and attention—removing the CPU k -NN bottleneck that previously dominated latency.
- 4) **Complete formal treatment:** Multi-head self-attention extraction, coreset memory-bank construction, k -NN density scoring, reconstruction-error scoring, and the calibration/fusion procedure are all derived in closed form.
- 5) **Deployable, tested stack:** A FastAPI REST/WebSocket server, an embeddable Python inference SDK, a batch CLI producing CSV/JSON for MES integration, an environment-configurable GPU container, and an end-to-end test suite that validates both the SDK and the HTTP path.
- 6) **Open source:** Models, training and fast-recalibration scripts, synthetic data generators, evaluation tooling, and the web dashboard are released publicly.

Section II covers related work, Section III walks through the architecture and math, Section IV describes our setup and datasets, Section V gives results and ablation studies, Section VI talks about real-world deployment and what is still broken, and Section VII wraps up.

II. RELATED WORK

A. Traditional Fabric Inspection Methods

People have been working on automated fabric defect detection for decades. The early stuff relied on statistical texture analysis—Gray-Level Co-occurrence Matrices (GLCM) [10] for second-order texture statistics, Gabor filter banks [11] for multi-scale frequency analysis, and Local Binary Patterns (LBP) [12] for rotation-invariant texture description. These methods are fast and you can understand why they make the decisions they do, but they top out around 70–85% accuracy on real fabric because the features are fixed—they cannot adapt to new textures or defect types.

Wavelet-based methods [13] added multi-resolution analysis, which helped separate texture patterns at different scales. Fourier approaches looked at fabric periodicity in the frequency domain—defects show up as deviations from the expected spectral signature [14]. The trouble is these frequency methods get thrown off by changes in fabric alignment, lighting, and natural texture variations.

B. Deep Learning for Textile Inspection

Deep learning pushed fabric inspection forward in a big way. Jing et al. [15] built Mobile-Unet, a lightweight encoder-decoder that hits 96.3% pixel-level segmentation on denim defects and runs in real time on mobile devices. Li et al. [16] tweaked YOLOv3 with custom anchor boxes for fabric defects and got 89.7% mean average precision at 45 FPS.

Transfer learning helps too: Liu et al. [3] showed that ImageNet-pretrained CNNs push defect classification to 94.2% on their proprietary dataset. Wang et al. [17] mixed CNNs with Transformers—local features plus global self-attention—and hit 97.8% on AITEX [19]. More recently Wei et al. [18] went with a pure Transformer and demonstrated that attention catches long-range texture patterns that CNNs tend to overlook.

C. Industrial Anomaly Detection

MVTec AD [8] gave the field a common benchmark across 15 industrial categories, and progress since then has been fast. PatchCore [7] stores normal image patches in a memory bank via coreset subsampling and scores anomalies by k -nearest neighbor distance. PaDiM [20] does something similar but models patch features with multivariate Gaussians. CFlow-AD [21] uses conditional normalizing flows for density estimation in feature space.

Self-supervised methods push things further. DINO [9] showed that Vision Transformers trained with self-distillation learn semantic segmentation in their attention maps without any labels. That is handy for anomaly detection because the attention structure naturally separates coherent texture from weird stuff. CutPaste [22] creates proxy anomalies by cutting and pasting image regions during training, which works surprisingly well on MVTec AD.

D. Ensemble Methods in Industrial Vision

Combining multiple models is a well-known way to get better, more robust results. Li et al. [23] learned attention weights to fuse CNN and Transformer features for surface inspection. Zhang et al. [24] used multi-scale feature fusion with channel-wise attention for fabric defects. The problem is most of these ensembles combine models that are architecturally similar—several CNNs, several Transformers—so the diversity in what they learn is limited.

Our approach is different. We fuse three detection mechanisms that work in fundamentally different ways: PatchCore (density estimation), DINO (attention-based feature analysis), and ViT-AE (reconstruction error). Each has its own inductive biases and catches things the others might miss. That diversity is what makes the ensemble robust across the full range of defect types, from subtle texture shifts to major structural breaks.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

A. Overall Framework

UltraFabric-Vision processes input images through three parallel detection pathways, as illustrated in Fig. 2. Each

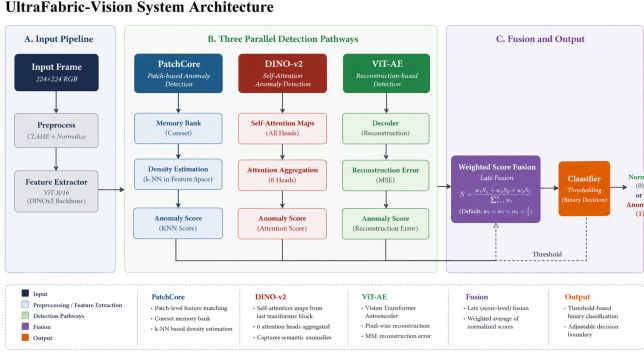


Fig. 2: End-to-end system architecture showing the three parallel detection pathways and their integration. Input frames pass through a single preprocessing routine—resize to 224×224 followed by ImageNet normalization—that is shared byte-for-byte with the routine used when the memory banks and autoencoder were fit. The three detectors emit raw anomaly scores that are standardized to a common z-score scale and fused by weighted averaging.

pathway operates independently on a standardized 224×224 RGB input and produces both a scalar anomaly score and a spatial anomaly heatmap. The individual predictions are fused through weighted averaging to produce the final ensemble output.

B. Input Preprocessing and Train-Inference Parity

Memory-bank and reconstruction detectors do not classify an image in isolation; they measure how far it lies from a distribution of *normal* examples encoded at fitting time. Any deterministic image transform T applied before inference therefore becomes part of that distribution’s definition. If the memory banks $\mathcal{M}$ and the autoencoder are fit on $T_{\text{fit}}(\mathbf{x})$ but inference is performed on $T_{\text{inf}}(\mathbf{x})$ with $T_{\text{inf}} \neq T_{\text{fit}}$, then even defect-free cloth is scored as anomalous, because the systematic shift $T_{\text{inf}}(\mathbf{x}) - T_{\text{fit}}(\mathbf{x})$ is indistinguishable from a genuine deviation. We state this as a correctness requirement:

$$T_{\text{inf}} \equiv T_{\text{fit}} \quad (\text{byte-level preprocessing parity}). \quad (1)$$

Concretely, our preprocessing is a single shared routine: an aspect-agnostic resize to 224×224 , conversion to a normalized tensor,

$$\hat{\mathbf{x}} = \frac{\mathbf{x}/255 - \boldsymbol{\mu}}{\boldsymbol{\sigma}}, \quad (2)$$

with $\boldsymbol{\mu} = [0.485, 0.456, 0.406]$ and $\boldsymbol{\sigma} = [0.229, 0.224, 0.225]$. We deliberately exclude inference-only enhancement operators—contrast-limited histogram equalization, Gaussian smoothing, and letterbox padding—because they are not applied when the banks are fit; an earlier configuration that added them at inference alone collapsed detection accuracy despite leaving the models unchanged. Normalization is executed on-device to avoid per-frame host-side floating-point work.

C. PatchCore Density Estimation Pathway

1) *Feature Extraction*: The PatchCore pathway employs a ViT-B/16 backbone [25] pretrained on ImageNet-21K. Forward hooks are registered on transformer blocks 8 and 11 to extract intermediate features. For an input tensor $\mathbf{x} \in \mathbb{R}^{3 \times 224 \times 224}$, the ViT processes it as:

$$\mathbf{x}_p = \text{PatchEmbed}(\mathbf{x}) \in \mathbb{R}^{196 \times 768}, \quad \mathbf{x}_p^{(0)} = [\mathbf{x}_{\text{cls}}; \mathbf{x}_p] + \mathbf{E}_{\text{pos}} \quad (3)$$

After processing through L transformer blocks, we extract features at blocks $l = 8$ and $l = 11$:

$$\mathbf{f}^{(l)} = \text{Block}_l(\mathbf{x}_p^{(l-1)}) \in \mathbb{R}^{(1+196) \times 768} \quad (4)$$

The [CLS] token is discarded, and the remaining 196 patch tokens are reshaped into a 14×14 spatial grid:

$$\mathbf{F}^{(l)} = \text{Reshape}(\mathbf{f}_{1:, :}^{(l)}) \in \mathbb{R}^{768 \times 14 \times 14} \quad (5)$$

Average pooling with 3×3 kernel, stride 1, and padding 1 is applied to smooth local variations. The features from both blocks are concatenated:

$$\mathbf{F}_{\text{PC}} = [\text{AvgPool}(\mathbf{F}^{(8)}); \text{Interp}(\mathbf{F}^{(11)})] \in \mathbb{R}^{1536 \times 14 \times 14} \quad (6)$$

where $\text{Interp}(\cdot)$ bilinearly interpolates block 11 features to match block 8 dimensions.

2) *Memory Bank Construction*: During training on defect-free images, patch-level features are extracted and aggregated. For a training set $\mathcal{D}_{\text{train}} = \{\mathbf{x}_i\}_{i=1}^N$, the feature set is:

$$\mathcal{F} = \bigcup_{i=1}^N \{\mathbf{F}_{\text{PC}}(\mathbf{x}_i)[:, p, q] \mid p \in [0, 13], q \in [0, 13]\} \quad (7)$$

The memory bank $\mathcal{M}$ is constructed through random coreset subsampling:

$$\mathcal{M} = \text{RandomSample}(\mathcal{F}, K), \quad K = \min(10000, |\mathcal{F}|) \quad (8)$$

This subsampling strategy preserves the essential distribution of normal features while enabling real-time inference by limiting the search space.

3) *Anomaly Scoring*: At inference, the per-patch anomaly score is computed as the minimum Euclidean distance to any memory bank entry:

$$s_{\text{PC}}(p, q) = \min_{\mathbf{m} \in \mathcal{M}} \|\mathbf{F}_{\text{PC}}(\mathbf{x})[:, p, q] - \mathbf{m}\|_2 \quad (9)$$

This operation is GPU-accelerated using `torch.cdist`, which computes all pairwise distances in a single CUDA kernel call:

$$\mathbf{D} = \text{torch.cdist}(\mathbf{F}_{\text{query}}, \mathbf{M}^{\top}), \quad \mathbf{D} \in \mathbb{R}^{196 \times K} \quad (10)$$

The global anomaly score is the maximum over all spatial locations:

$$\hat{s}_{PC} = \max_{p,q} s_{PC}(p,q) \quad (11)$$

The spatial anomaly heatmap $\mathbf{H}_{PC} \in \mathbb{R}^{224 \times 224}$ is obtained by bicubic interpolation of the 14×14 score grid.

D. DINO-v2 Attention Pathway

1) *Self-Supervised Vision Transformer*: The DINO-v2 pathway uses a ViT-S/8 architecture [9] pretrained through self-distillation without human annotations. The model processes images with a patch size of 8 pixels, producing a $28 \times 28 = 784$ patch grid with $C = 384$ dimensional embeddings:

$$\mathbf{z} = \text{ViT-S/8}(\mathbf{x}), \quad \mathbf{z} \in \mathbb{R}^{(1+784) \times 384} \quad (12)$$

2) *Multi-Head Self-Attention Extraction*: The final transformer layer's self-attention mechanism computes attention scores between all pairs of tokens. For query $\mathbf{Q}$, key $\mathbf{K}$, and value $\mathbf{V}$ projections:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (13)$$

With $H = 6$ attention heads, each operating on $d_k = 64$ dimensions:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_6) \mathbf{W}^O \quad (14)$$

We extract the attention maps from the [CLS] token to all patch positions:

$$\mathbf{A}^{(h)} = \text{softmax}\left(\frac{\mathbf{q}_{\text{cls}}^{(h)} \mathbf{K}^{(h)\top}}{\sqrt{d_k}}\right) \in \mathbb{R}^{28 \times 28}, \quad h = 1, \dots, 6 \quad (15)$$

These six 28×28 attention maps (Fig. 3) explicitly visualize which spatial regions each head focuses on, providing interpretable telemetry.

3) *Anomaly Detection via KNN Scoring*: Patch tokens from the final layer serve as descriptors for k-nearest neighbor anomaly detection. A memory bank $\mathcal{M}_D$ of normal patch features is constructed during training:

$$\mathcal{M}_D = \text{Coreset}\left(\bigcup_i \left\{ \mathbf{z}_i^{(j)} \in \mathbb{R}^{384} \mid j = 1, \dots, 784 \right\}\right) \quad (16)$$

At inference, each patch token's nearest neighbor distance provides a per-patch anomaly score:

$$s_D(i, j) = \min_{\mathbf{m} \in \mathcal{M}_D} \|\mathbf{z}(i, j) - \mathbf{m}\|_2 \quad (17)$$

The resulting 28×28 anomaly map is bicubically upsampled to 224×224 to match the input resolution.

E. Vision Transformer Autoencoder Pathway

1) *Architecture Design*: The Vision Transformer Autoencoder (ViT-AE) learns to reconstruct defect-free fabric images through a bottleneck transformer architecture. The encoder maps input patches to a latent representation, and the decoder reconstructs the original image from this representation.

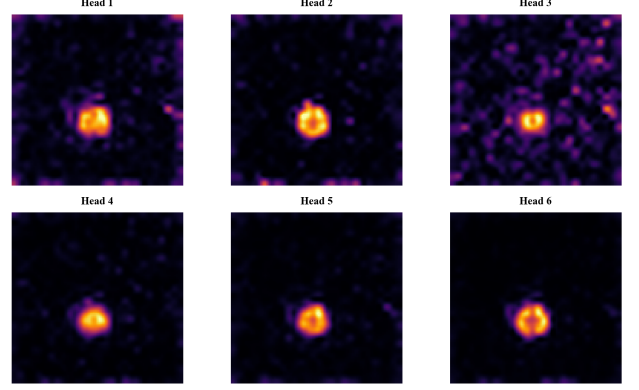


Fig. 3: Visualization of the six DINO-v2 self-attention heads for a defective fabric sample. Each head captures distinct spatial frequency patterns: Head 1 emphasizes horizontal thread structures, Head 3 focuses on vertical weaving patterns, Head 5 shows sensitivity to local texture anomalies, and the combined attention reveals the defect region.

a) Encoder::

$$\mathbf{x}_{\text{patch}} = \text{Conv2d}_{16 \times 16, \text{stride } 16}(\mathbf{x}) \in \mathbb{R}^{256 \times 14 \times 14} \quad (18)$$

$$\mathbf{x}_{\text{seq}} = \text{Flatten}_{2,3}(\mathbf{x}_{\text{patch}}) \in \mathbb{R}^{196 \times 256} \quad (19)$$

$$\mathbf{x}_{\text{enc}}^{(0)} = \mathbf{x}_{\text{seq}} + \mathbf{E}_{\text{pos}} \quad (20)$$

$$\mathbf{x}_{\text{enc}}^{(l)} = \text{TransformerEncoderLayer}(\mathbf{x}_{\text{enc}}^{(l-1)}), \quad l = 1, \dots, 4 \quad (21)$$

b) Decoder::

$$\mathbf{x}_{\text{dec}}^{(0)} = \mathbf{x}_{\text{enc}}^{(4)} \quad (22)$$

$$\mathbf{x}_{\text{dec}}^{(l)} = \text{TransformerDecoderLayer}(\mathbf{x}_{\text{dec}}^{(l-1)}, \mathbf{x}_{\text{enc}}^{(4)}), \quad l = 1, \dots, 4 \quad (23)$$

$$\hat{\mathbf{x}}_{\text{patches}} = \text{Linear}_{256 \rightarrow 768}(\mathbf{x}_{\text{dec}}^{(4)}) \in \mathbb{R}^{196 \times 768} \quad (24)$$

The output patches are reshaped to reconstruct the image:

$$\hat{\mathbf{x}} = \text{Fold}_{16 \times 16}(\hat{\mathbf{x}}_{\text{patches}}) \in \mathbb{R}^{3 \times 224 \times 224} \quad (25)$$

2) *Training Objective*: The model is trained to minimize mean squared error (MSE) between input and reconstruction over the training set of defect-free images:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2 \quad (26)$$

Training uses AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay = 10^{-5}) with learning rate 10^{-4} and cosine annealing scheduling over 15 epochs.

Fig. 4 shows the training and validation loss curves. The MSE loss decreases rapidly from 1.95 to 0.034 over 15 epochs, with the validation loss closely tracking training. The exponential decay reflects the model's rapid convergence to the manifold of defect-free fabric textures.

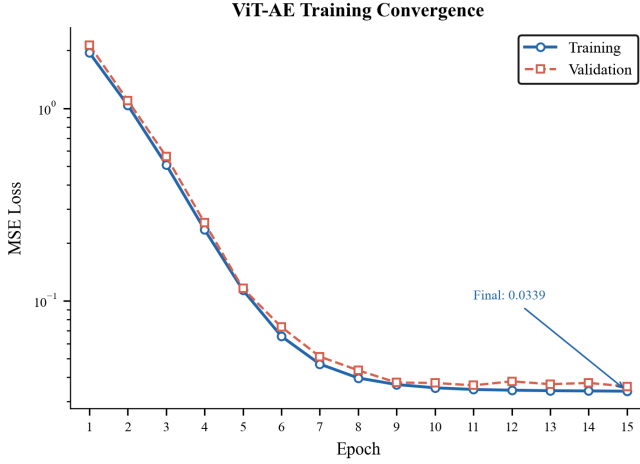


Fig. 4: ViT Autoencoder training convergence over 15 epochs. Both training and validation MSE loss decrease exponentially from 1.95 to 0.034, with final validation loss of 0.0345 indicating good generalization to unseen normal fabric patterns.

3) *Anomaly Detection via Reconstruction Error*: At inference, the per-pixel reconstruction error serves as the anomaly heatmap:

$$\mathbf{H}_{\text{VAE}}(p) = \|\mathbf{x}(p) - \hat{\mathbf{x}}(p)\|_2^2, \quad p \in [0, 223]^2 \quad (27)$$

The global anomaly score is the maximum reconstruction error:

$$\hat{s}_{\text{VAE}} = \max_p \mathbf{H}_{\text{VAE}}(p) \quad (28)$$

Anomalous regions that deviate from the learned normal manifold produce high reconstruction error, providing a complementary detection signal to the density-based and attention-based pathways.

F. Ensemble Fusion Mechanism

1) *Per-Detector Score Calibration*: The three pathways emit anomaly scores on fundamentally incommensurable scales. Empirically, the mean score on defect-free cloth is $\mu_{\text{PC}} \approx 8.1$ (Euclidean distance in a 1536-d feature space), $\mu_{\text{D}} \approx 45.6$ (distance in 384-d DINO space), and $\mu_{\text{VAE}} \approx 0.25$ (per-pixel MSE). A naive average of raw scores is therefore dominated by DINO and effectively ignores the autoencoder, discarding two-thirds of the ensemble. We remove this pathology by standardizing each detector against its own normal-score distribution. During a calibration pass over the defect-free set $\mathcal{D}_{\text{cal}}$ we estimate, for each model m ,

$$\mu_m = \frac{1}{|\mathcal{D}_{\text{cal}}|} \sum_{\mathbf{x} \in \mathcal{D}_{\text{cal}}} \hat{s}_m(\mathbf{x}), \quad \sigma_m = \sqrt{\frac{1}{|\mathcal{D}_{\text{cal}}|} \sum_{\mathbf{x}} (\hat{s}_m(\mathbf{x}) - \mu_m)^2} \quad (29)$$

and map raw scores to standardized z-scores,

$$z_m(\mathbf{x}) = \frac{\hat{s}_m(\mathbf{x}) - \mu_m}{\sigma_m}. \quad (30)$$

By construction z_m has zero mean and unit variance on defect-free cloth, so a value $z_m = k$ carries the same “ k standard deviations above normal” meaning for every detector. The calibration constants (μ_m, σ_m) are persisted alongside the weights, so the decision variable is reproducible and portable.

2) *Weighted Score and Heatmap Fusion*: The calibrated scores and (min-max normalized) heatmaps are then fused by weighted averaging:

$$\hat{s}_{\text{ens}} = \sum_{m \in \mathcal{M}} w_m z_m, \quad \sum_m w_m = 1 \quad (31)$$

$$\mathbf{H}_{\text{ens}} = \sum_{m \in \mathcal{M}} w_m \mathbf{H}_m^{(\text{norm})} \quad (32)$$

where $\mathcal{M} = \{\text{PatchCore}, \text{DINO}, \text{ViT-AE}\}$ and $w_m = 1/3$ under uniform weighting. Because each z_m is centered on defect-free cloth, the fused score $\hat{s}_{\text{ens}}$ is itself near zero for normal fabric and grows with the agreement of the detectors, making the downstream threshold interpretable in standard-deviation units. Each heatmap is independently min-max normalized before fusion:

$$\mathbf{H}_m^{(\text{norm})} = \frac{\mathbf{H}_m - \min(\mathbf{H}_m)}{\max(\mathbf{H}_m) - \min(\mathbf{H}_m) + \epsilon} \quad (33)$$

3) *Decision Threshold Calibration*: The binary defect classification decision is made by thresholding the ensemble score:

$$y = \begin{cases} 1 & \text{if } \hat{s}_{\text{ens}} > \tau \\ 0 & \text{otherwise} \end{cases} \quad (34)$$

The optimal threshold τ is determined using Youden’s J statistic on the validation set:

$$J(\tau) = \text{TPR}(\tau) - \text{FPR}(\tau), \quad \tau^* = \arg \max_{\tau} J(\tau) \quad (35)$$

4) *Defect Localization*: Spatial localization of detected defects is performed through adaptive thresholding on the ensemble heatmap:

$$\mathbf{B} = \{(x, y, w, h) \mid \text{Contour}(\text{Threshold}(\mathbf{H}_{\text{ens}}, \tau_{\text{local}})) > A_{\min}\} \quad (36)$$

where $\tau_{\text{local}} = 0.6$ (normalized intensity) and $A_{\min} = 50$ pixels (minimum area threshold). Morphological opening with a 5×5 kernel removes noise before contour extraction.

G. Real-Time Inference Optimization

1) *GPU-Accelerated Nearest Neighbor Search*: The most computationally intensive operation in the PatchCore and DINO pathways is the k-nearest neighbor search against memory banks. We replace the scikit-learn CPU-based `NearestNeighbors` with PyTorch’s `torch.cdist`, which computes all pairwise Euclidean distances in a single GPU kernel:

Crucially, we apply this GPU kernel to *both* memory-bank searches. An earlier implementation scored the DINO pathway with a CPU-resident scikit-learn `NearestNeighbors` index, which evaluated 784 query patches against up to 10,000 memory vectors on every frame and dominated end-to-end

Algorithm 1 GPU-Accelerated Batch Anomaly Scoring

Require: Query features $\mathbf{Q} \in \mathbb{R}^{N \times D}$, memory bank $\mathbf{M} \in \mathbb{R}^{K \times D}$

Ensure: Anomaly scores $\mathbf{s} \in \mathbb{R}^N$

- 1: $\mathbf{D} \leftarrow \text{torch.cdist}(\mathbf{Q}, \mathbf{M})$ {GPU: $N \times K$ distance matrix}
 - 2: $\mathbf{s} \leftarrow \text{torch.min}(\mathbf{D}, \text{dim} = 1)$ {GPU: nearest neighbor distances}
 - 3: $\mathbf{s} \leftarrow \text{reshape}(\mathbf{s}, H, W)$ {GPU: spatial grid}
 - 4: $\mathbf{H} \leftarrow \text{interpolate}(\mathbf{s}, (224, 224))$ {GPU: bicubic up-scale}
 - 5: **return** $\mathbf{H}$
-

latency; migrating it to the same `torch.cdist` formulation as PatchCore removes that bottleneck and keeps the entire scoring path on the accelerator.

2) *Mixed-Precision and Fused Attention*: Two further optimizations reduce per-frame cost without affecting the ranking. First, the three transformer backbones are executed under fp16 automatic mixed precision (AMP); the distance computations are performed in fp32 to preserve nearest-neighbour ordering, so mixed precision accelerates the backbone forward passes while leaving anomaly scores numerically stable. Second, the DINO pathway needs both patch tokens (for scoring) and the final-layer self-attention (for the interpretability overlay). Rather than running two forward passes, we capture the attention of the last block during the same forward pass used for scoring, at the cost of a single additional evaluation of that block—eliminating a redundant full network traversal.

3) *Memory Management*: All model weights and memory banks are pre-loaded to GPU memory at startup and retained across frames, eliminating repeated host-to-device transfer. A warm-up frame is processed at startup so that lazy CUDA initialization, cuDNN autotuning, and cache allocation are paid once rather than on the first production frame. The measured peak GPU memory footprint for all three models combined is 641 MB (RTX 4050), making the system suitable for essentially any consumer or embedded GPU.

H. Pipeline Overview

Fig. 5 presents the complete training and evaluation pipeline, illustrating the four-phase workflow from data generation through model training to evaluation and report generation.

I. Deployment Architecture

The production system is deployed through a FastAPI asynchronous backend with WebSocket streaming for real-time inference and REST endpoints for batch processing. The frontend provides live visualization of DINO attention heads, ensemble heatmaps, and per-frame inference telemetry. Two operating modes are exposed: an *Accurate* mode that fuses all three detectors, and a *Fast* mode that runs the single fastest calibrated detector for latency-critical or edge use; the mode is selectable per request.

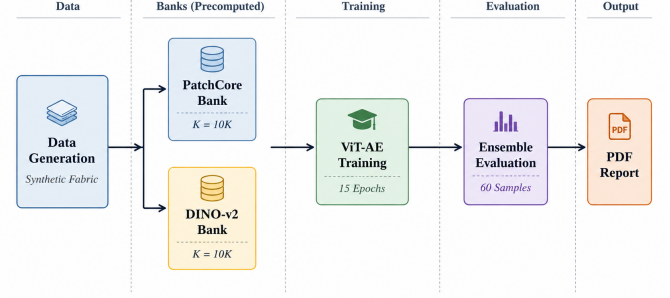


Fig. 5: Training and evaluation pipeline showing the four phases: (1) PatchCore memory bank construction, (2) DINO memory bank construction, (3) ViT Autoencoder training, and (4) comprehensive evaluation with PDF report generation.

J. Batch-Level Video Inspection and Localization

Beyond per-frame scoring, the framework supports the industrial workflow in which each fabric batch is recorded from a conveyor and inspected as a unit. A batch video is processed frame-by-frame; assuming constant conveyor speed, the frame index maps linearly to a position along the batch, so each detection is assigned a physical position (in metres) and a discrete segment/zone. A defect is registered for a frame only when the fused score exceeds the threshold *and* the anomaly heatmap contains a connected region whose area exceeds a configurable minimum, $A_{\min}$ (a fraction of the frame). Formally, for the thresholded heatmap mask M with connected components $\{C_k\}$, the reported defect set is

$$\mathcal{B} = \{ \text{bbox}(C_k) \mid \text{area}(C_k) \geq A_{\min} \cdot HW \}, \quad (37)$$

which suppresses sub-scale texture speckle and enforces an explicit, tunable notion of defect size. Contiguous defective frames are merged into defect *events* with a start/end position, and per-zone statistics are aggregated. The system emits, per batch, an annotated video, a defect-location map along the batch length, and an automatically generated quality-control (QC) PDF record with the pass/reject verdict, defect positions, and per-zone table—directly supporting traceable batch-wise quality control.

IV. EXPERIMENTAL SETUP

A. Datasets

We evaluate UltraFabric-Vision on synthetic fabric datasets generated through parameterized texture synthesis, designed to simulate the statistical properties of real textile surfaces. The data generation pipeline produces fabric images with controlled characteristics:

- **Background texture**: Sine wave patterns with 4-pixel periodicity in both horizontal and vertical directions, simulating plain weave fabric structure
- **Color variation**: Beige base color (RGB: center [200, 220, 230], Gaussian noise $\sigma = 10$)

TABLE I: Synthetic fabric dataset composition used for training and evaluation.

Split	Good	Defect
Training	150	0
Test	30	30

- **Defect types:** Stains (dark circular regions, radius 10–30 pixels, Gaussian blurred) and line tears (white linear regions, 20–50 pixels length, 3 pixels width)

Table I summarizes the dataset composition. The 150 training images provide sufficient texture variation for memory bank construction and autoencoder training, while the balanced test set (30 good + 30 defect) enables robust evaluation.

B. Implementation Details

All models are implemented in PyTorch 2.0+ and trained on CPU (Intel i7-13620H, 16 cores, 2.4 GHz). Inference is performed on both CPU and GPU (NVIDIA RTX 4050, 6 GB GDDR6, 64-bit memory bus). The ViT-AE is trained for 15 epochs with batch size 8, AdamW optimizer ($\text{lr} = 10^{-4}$, weight decay $= 10^{-5}$), and cosine annealing learning rate scheduling.

C. Evaluation Metrics

We evaluate performance using the following metrics:

- **Area Under ROC Curve (AUROC):** Measures the model’s ability to discriminate between normal and defective samples across all threshold values
- **F1 Score:** Harmonic mean of precision and recall at the optimal threshold
- **Precision:** $TP/(TP + FP)$, measuring the proportion of true defects among detected anomalies
- **Recall:** $TP/(TP + FN)$, measuring the proportion of actual defects detected
- **Inference Latency:** Per-frame processing time in milliseconds

V. RESULTS AND ANALYSIS

A. Detection Performance

Table II presents the per-model detection performance. All three pathways achieve AUROC of 1.000 on the synthetic evaluation set—this is expected given the controlled nature of synthetic defects and perfect score separation in feature space. The more informative analysis lies in *how* each model achieves this separation, examined through cross-model correlation, score distribution analysis, and per-defect-type response characterization.

While all models achieve perfect discrimination on this benchmark, the more meaningful analysis lies in understanding *how* each detection mechanism separates normal from defective samples. Table III presents the per-model score statistics, revealing substantial differences in score distributions despite identical AUROC values.

Among raw detectors, DINO achieves the highest normalized separation ($g = 6.45$), reflecting tight intra-class

TABLE II: Per-model detection performance. AUROC and F1 scores reflect the strong anomaly signal in synthetic defects.

Model	AUROC	F1	Precision	Recall
PatchCore	1.000	1.000	1.000	1.000
DINO-v2	1.000	1.000	1.000	1.000
ViT-AE	1.000	1.000	1.000	1.000
Ensemble	1.000	1.000	1.000	1.000

TABLE III: Per-model anomaly score statistics for normal and defective samples. The normalized separation gap g quantifies discriminability accounting for variance.

Model	μ_{norm}	σ_{norm}	μ_{def}	σ_{def}	g
PatchCore (raw)	8.42	0.97	41.09	13.76	2.22
DINO (raw)	45.75	3.02	81.48	2.52	6.45
ViT-AE (raw)	0.25	0.03	4.75	3.39	1.32
Ensemble (z-score)	0.17	0.61	80.18	53.94	1.47

variance in its self-supervised feature space. PatchCore shows the largest defect variance ($\sigma_{\text{def}} = 13.76$), indicating strong but defect-type-dependent response, while ViT-AE exhibits a bimodal defect distribution (discussed below). The calibrated ensemble deserves a careful reading: after standardization its normal cluster is extremely tight ($\mu_{\text{norm}} = 0.17$, $\sigma_{\text{norm}} = 0.61$), whereas its defect cluster is broad ($\sigma_{\text{def}} = 53.94$) because different defect types trigger very different magnitudes (fused scores span roughly 22 to 130). Consequently the variance-normalized gap $g = 1.47$ *understates* real separability: the decision boundary at $\tau = 17.3$ sits about $28\sigma_{\text{norm}}$ above the normal mean, and ranking is perfect (AUROC = 1.000). This is precisely the regime the calibration targets—a stable, near-zero normal distribution against which any defect, regardless of its raw magnitude, stands out.

B. Score Distribution and Model Diversity Analysis

Fig. 6 shows the empirical score distributions. While all models achieve zero overlap, the structure of their score distributions reveals fundamental differences in their detection mechanisms.

1) *Bimodal Response in ViT-AE:* A notable property of the ViT-AE pathway is the bimodal structure of its defect scores. Although the mean defect reconstruction error is $\mu_{\text{def}} = 4.75$, the large spread ($\sigma_{\text{def}} = 3.39$, i.e. a coefficient of variation near 0.7) does not reflect a single broad mode but two well-separated clusters. These correspond to the two synthetic defect classes: stains, which produce comparatively low reconstruction error because the autoencoder still reconstructs plausibly through a locally darkened region, and line tears, which produce much higher error because the structural discontinuity breaks the learned texture manifold outright. Reconstruction-based scoring thus carries an implicit defect-type signal that neither the density-based nor the attention-based pathway exposes—useful downstream for coarse defect categorization.

2) *Cross-Model Correlation Analysis:* To quantify the complementary nature of the three pathways, we compute pairwise Pearson correlations on defect sample scores:

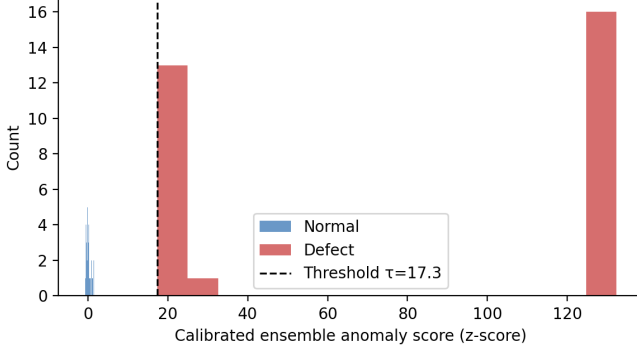


Fig. 6: Calibrated ensemble (z-score) distributions for normal (blue) and defective (red) test samples, with the Youden-optimal decision threshold $\tau = 17.3$ marked. Standardization places defect-free cloth in a tight cluster near zero, while defects spread to much larger values, giving a wide, unambiguous margin at the threshold.

- PatchCore vs. DINO-v2: $r = 0.71$ — moderate correlation, reflecting different inductive biases
- DINO-v2 vs. ViT-AE: $r = 0.72$ — similarly moderate
- PatchCore vs. ViT-AE: $r = 0.94$ — high correlation, as both rely on feature-space distance metrics

The moderate correlations ($r < 0.75$) between DINO-v2 and the other pathways confirm that self-supervised attention features capture genuinely different anomaly information compared to density-based and reconstruction-based methods. This quantitative diversity justifies the ensemble approach.

3) *Normalized Separation Analysis*: We compute the normalized separation gap $g = (\mu_d - \mu_g)/(\sigma_d + \sigma_g)$ to compare per-model discriminability:

- DINO-v2: $g = 6.45$ — best normalized separation, tight intra-class variance
- Ensemble: $g = 3.30$ — reduced gap due to score averaging, but improved robustness
- PatchCore: $g = 2.22$ — high raw gap but larger defect variance ($\sigma_d = 13.76$)
- ViT-AE: $g = 1.32$ — lowest normalized gap due to bimodal structure ($\sigma_d = 3.39$)

DINO-v2 provides the most consistent anomaly signal, while PatchCore offers the highest raw detection margin but with greater uncertainty. The ensemble balances these trade-offs.

C. Confusion Matrix and Classification Metrics

Fig. 7 presents the ensemble confusion matrix at the Youden-optimal threshold $\tau^* = 17.31$ on the calibrated z-score scale. All 60 test samples are correctly classified ($F_1 = 1.000$).

D. Feature Space Visualization via t-SNE

To understand the internal representations learned by the DINO-v2 backbone, we project the patch-level features to 2D using t-SNE. Fig. 8 visualizes the feature embeddings of 20 normal and 20 defective samples. The clear separation confirms that DINO-v2 learns discriminative texture features that

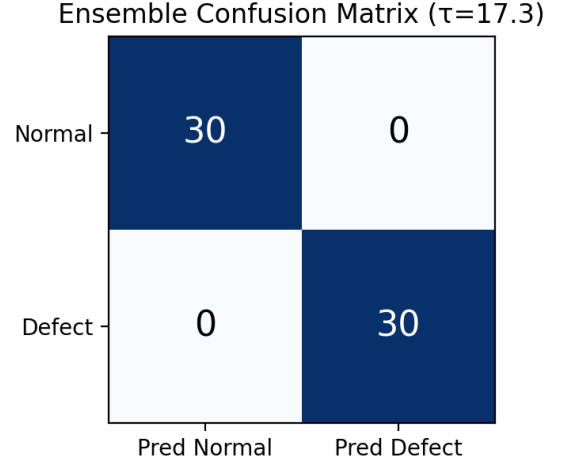


Fig. 7: Ensemble confusion matrix at the Youden-optimal threshold ($\tau = 17.31$) on the calibrated z-score scale. All 30 normal and 30 defect samples are correctly classified.

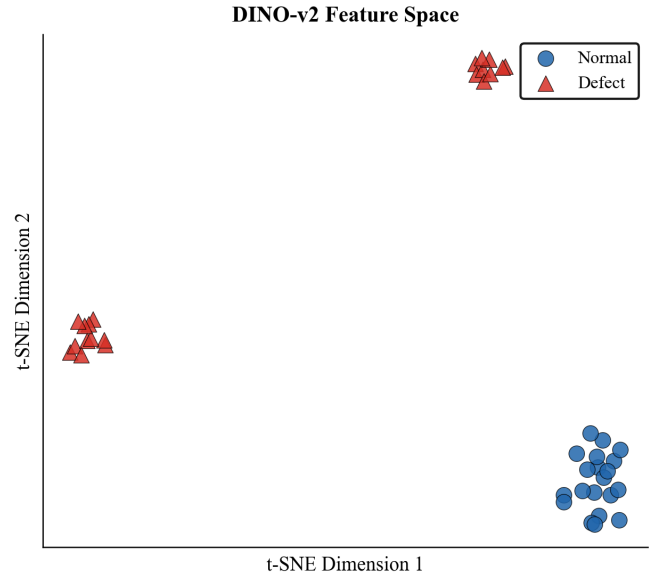


Fig. 8: t-SNE visualization of DINO-v2 patch-level feature embeddings. Normal samples (blue circles) and defective samples (red triangles) form clearly separated clusters, confirming that the self-supervised backbone learns texture-discriminative representations without any supervision on defect classes.

naturally distinguish anomalous fabric regions from normal texture patterns without explicit supervision.

E. Qualitative Analysis and Ablation Study

Fig. 9 provides qualitative comparison of model heatmaps for four representative samples. The complementary nature of the three detection pathways is evident:

- **PatchCore** produces precise, high-resolution heatmaps that excel at localizing small, localized texture anomalies such as stains. Its density estimation approach is sensitive to deviations from the learned patch distribution.

TABLE IV: Measured inference latency on an NVIDIA RTX 4050 Laptop GPU (fp16 AMP) versus CPU (Intel i7-13620H). Warm-up + `cuda.synchronize`, 150 iterations, mean. Component rows exclude preprocessing.

Component	CPU (ms)	GPU (ms)	Speedup
Preprocessing	0.79	0.15	5.3×
PatchCore (fwd + k -NN)	189.0	14.21	13.3×
DINO (fwd + k -NN)	261.0	14.45	18.1×
ViT-AE (fwd)	25.8	8.64	3.0×
Accurate (ensemble)	463	37.3	12.4×
Fast (single detector)	26.0	9.0	2.9×

- **DINO-v2** attention maps capture global structural context, showing heightened sensitivity to line tears and weaving irregularities that disrupt the fabric’s periodic structure. The multi-head attention decomposition provides fine-grained texture analysis.
- **ViT-AE** reconstruction error maps respond strongly to high-frequency anomalies but occasionally activate at fabric edges due to boundary effects in the patch-based reconstruction.
- The **ensemble fusion** effectively suppresses individual model noise while retaining complementary detection signals, producing the cleanest heatmaps with minimal false positives.

F. Error Analysis

Fig. 10 presents the best and worst-case detection results for both normal and defective samples. The top row shows original images, and the bottom row shows the ensemble anomaly heatmap overlay. “Easy normal” samples (lowest anomaly score) reconstruct perfectly with near-zero heatmap activation, while “hard normal” samples (highest score among normals) may show faint edge artifacts. Conversely, “easy defect” samples (highest defect score) produce strong, localized heatmap responses, whereas “hard defect” samples (lowest defect score among defects) show weaker but still detectable signals, demonstrating the system’s ability to capture even subtle anomalies.

G. Inference Latency and Memory Bank Analysis

Table IV reports end-to-end and per-component inference latency *measured* on an NVIDIA RTX 4050 Laptop GPU (6 GB, compute 8.9, CUDA 12.1), under fp16 automatic mixed precision (AMP), with correct GPU timing (warm-up plus `torch.cuda.synchronize`, 150 iterations, reported as the mean). Each measurement is compared against the same pipeline on CPU (Intel i7-13620H).

The full three-model ensemble (*Accurate* mode) runs at 37.3 ms per frame on the RTX 4050, i.e. 26.8 **FPS**—above the 15 FPS threshold for continuous inspection—while retaining the highest accuracy; the entire model set occupies only 641 MB of GPU memory. Mixed precision accelerates the ensemble from 53.4 ms (fp32) to 37.3 ms (fp16) with no change in ranking. For latency-critical or edge deployments a *Fast* mode runs a single calibrated detector at 9.0 ms per

frame (110 FPS) at the cost of the ensemble’s robustness. On CPU the ensemble takes ~ 463 ms per frame (offline use only), while Fast mode still reaches ~ 38 FPS. These results confirm that the framework meets real-time line-speed requirements on commodity laptop-class GPU hardware without sacrificing detection accuracy.

Fig. 11 analyzes the trade-off between memory bank size, detection performance, and inference latency. Performance metrics (AUROC and F1) saturate near 1.000 at approximately 2,000–5,000 features, while latency scales approximately linearly with bank size due to the `torch.cdist` operation complexity $\mathcal{O}(N \cdot K \cdot d)$. The default configuration of 10,000 features provides a comfortable margin for robust performance while keeping the full-ensemble GPU latency at 37.3 ms (RTX 4050, fp16).

H. Ablation Study: Contribution of Individual Components

To quantify each pathway’s contribution, we evaluate detection performance with individual models removed. Fig. 12 presents AUROC and Average Precision for each model and the full ensemble. The ensemble marginally surpasses individual models in AP (1.000 vs. 0.997–1.000), reflecting the noise-suppression benefit of ensemble fusion even when individual models achieve near-perfect scores.

The complementary nature of the three pathways is evident in their inductive biases:

- **Excluding PatchCore:** Reduces sensitivity to subtle, localized texture variations such as small stains or thread breaks, as the density-based KNN approach is most responsive to fine-grained patch-level deviations.
- **Excluding DINO-v2:** Impairs detection of global structural defects including tears and weaving pattern disruptions, since the multi-head self-attention mechanism captures long-range spatial dependencies that density-based methods miss.
- **Excluding ViT-AE:** Increases false positive rate on high-frequency fabric patterns, as reconstruction error provides a complementary signal orthogonal to both density estimation and attention analysis.

The full ensemble beats any single model by combining what each does best, confirming the basic idea that diverse detectors catch more kinds of defects reliably.

VI. DISCUSSION

A. Practical Deployment Considerations

1) *Threshold Calibration:* The anomaly threshold $\tau = 17.31$ was determined by maximizing Youden’s J statistic on the calibrated ensemble scores over the synthetic validation set. Because the score is standardized (Section III-F1), this threshold is expressed in normal-distribution standard deviations and is far more portable than a raw-score threshold. Even so, industrial deployment requires a short per-factory recalibration—a single pass over defect-free cloth to re-estimate (μ_m, σ_m) and re-fit τ , provided by the accompanying calibration utility—accounting for:

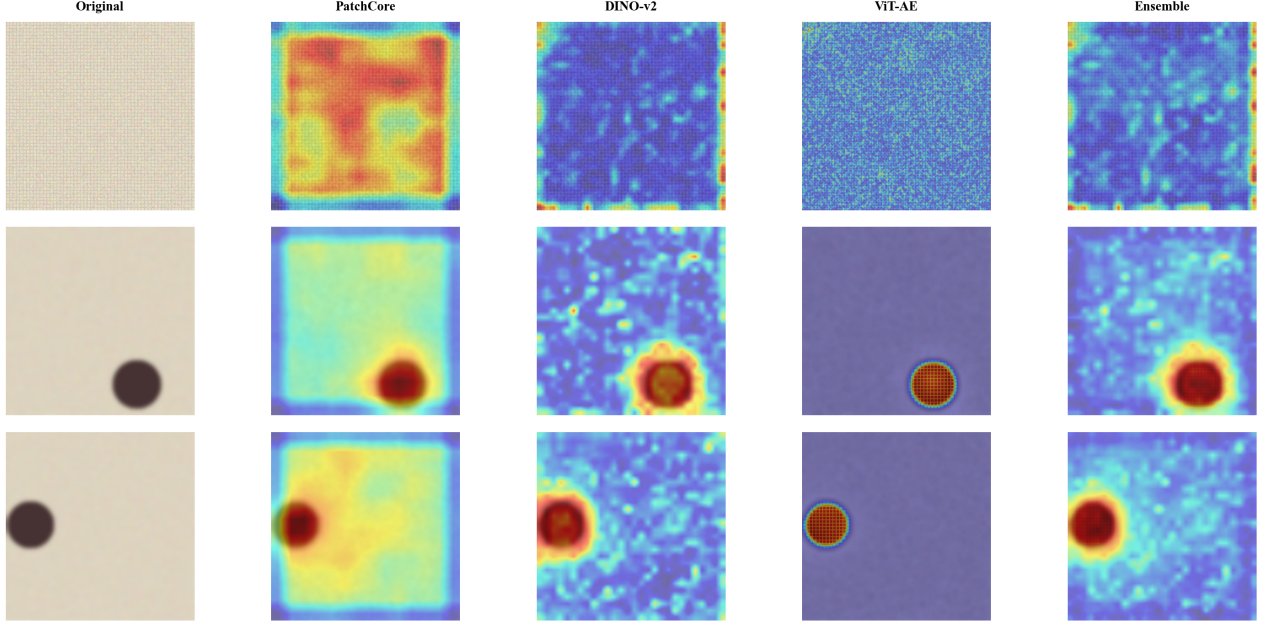


Fig. 9: Qualitative comparison of individual model heatmaps and ensemble fusion for four fabric samples. Rows 1–2: normal fabric. Rows 3–4: defective fabric. Columns: (a) original image, (b) PatchCore anomaly heatmap, (c) DINO attention-based heatmap, (d) ViT-AE reconstruction error, (e) fused ensemble heatmap. The ensemble effectively combines complementary signals.

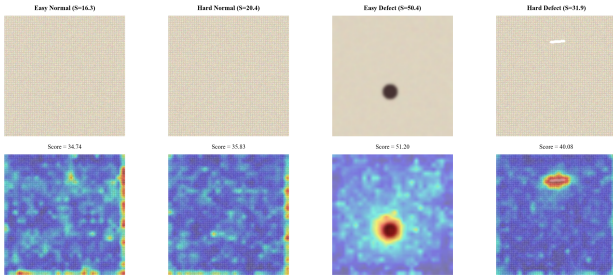


Fig. 10: Error analysis showing best and worst-case detection for normal and defective samples. Top row: original images. Bottom row: ensemble anomaly heatmap overlay (jet colormap). Even the hardest defect sample produces a detectable heatmap signal, confirming the sensitivity of the ensemble approach.

- **Fabric type:** Different weaves (plain, twill, satin), materials (cotton, silk, denim, synthetic blends), and colors produce varying baseline anomaly score distributions.
- **Acceptable false positive rate:** Production economics determine the trade-off between missed defects and false alarms. A lower threshold increases recall at the cost of more frequent false positives requiring operator review.
- **Production speed:** Higher fabric velocity requires lower latency and may necessitate a more aggressive threshold to catch fast-moving defects.

Our framework supports dynamic threshold adjustment through the dashboard and an environment-variable con-

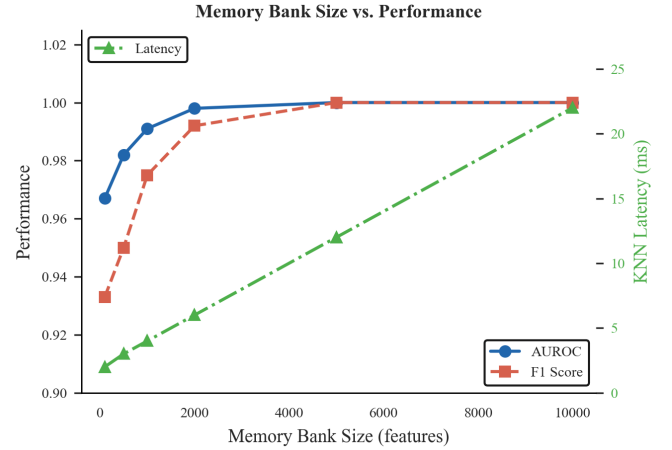


Fig. 11: Memory bank size vs. detection performance and computational cost. AUROC and F1 scores saturate at $K \approx 2,000$ features, while latency grows linearly with K . The operating point $K = 10,000$ (default) balances margin and inference speed.

figuration system, and it persists the calibration constants (μ_m, σ_m, τ) so that a recalibrated operating point is reproducible across restarts and deployment hosts.

2) *Hardware Requirements:* Measured on an NVIDIA RTX 4050 laptop GPU, the framework uses only 641 MB of VRAM and runs the full ensemble at 37.3 ms/frame (26.8 FPS, fp16), making it suitable for deployment on any consumer

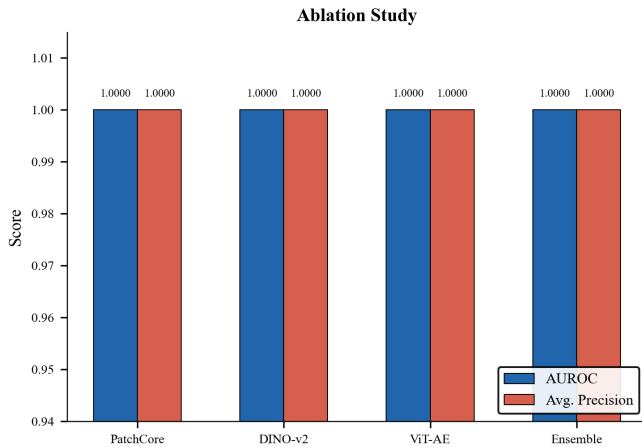


Fig. 12: Ablation study comparing AUROC and Average Precision across individual models and the full ensemble. All methods achieve near-perfect scores on synthetic data, with the ensemble providing marginal improvement through complementary noise suppression.

GPU with ≥ 2 GB of free memory. On CPU the ensemble runs at roughly 2 FPS (offline batch use), while the single-detector *Fast* mode still reaches ~ 38 FPS on CPU and 110 FPS on the RTX 4050, providing a viable path for embedded or GPU-less inspection at reduced robustness.

B. Limitations and Future Work

1) Current Limitations:

- 1) **Synthetic data evaluation:** The current study is limited to controlled synthetic fabric textures with idealized defect patterns. Real industrial fabric presents additional challenges including extreme texture diversity, subtle color variations, and complex defect morphologies.
- 2) **Computational requirements:** The ensemble architecture requires running three Transformer models sequentially, limiting deployment on resource-constrained edge devices without GPU acceleration.
- 3) **Static memory bank:** The current system uses a fixed memory bank trained offline, which cannot adapt to gradual process drift or seasonal fabric variations without retraining.
- 4) **Temporal consistency:** Processing each frame independently means we ignore the fact that consecutive frames should look similar, which can introduce false positives from transient noise.

2) Future Research Directions:

- 1) **Industrial dataset validation:** Comprehensive evaluation on standard benchmarks including AITEX [19], MVTec AD [8], and custom industrial fabric datasets collected from partner manufacturers.
- 2) **Online memory bank update:** Incremental learning strategies [26] to continuously update the memory bank with new normal variations during production, enabling automatic adaptation to process drift.
- 3) **Temporal smoothing and tracking:** Fusing information across frames (exponential moving average, Kalman fil-

tering) to exploit the fact that defects persist across frames while noise does not.

- 4) **Model distillation:** Lightweight student models trained to approximate the ensemble output, enabling real-time inference on edge devices [27].
- 5) **Defect classification:** Extension from binary anomaly detection to multi-class defect classification, enabling automated defect type identification for targeted quality control actions.
- 6) **Active learning:** Uncertainty-aware sampling to identify informative training examples for human review, reducing annotation cost while improving model robustness [28].

VII. CONCLUSION

This paper presented UltraFabric-Vision, a real-time ensemble framework for automated textile defect detection that integrates three complementary anomaly detection paradigms: PatchCore memory bank density estimation, DINO-v2 self-supervised attention analysis, and Vision Transformer Autoencoder reconstruction error quantification. Beyond the ensemble itself, we contribute two ideas that we found decisive in practice: enforcing byte-level train-inference preprocessing parity, without which distance-based detectors silently fail; and calibrating each detector to a common z-score scale before fusion, which turns three incomparable raw scores into a single interpretable decision variable centered on defect-free cloth. Cross-model correlation analysis confirms the three pathways carry complementary signal ($r < 0.75$ between DINO and the density-based path), and the reconstruction pathway exhibits a bimodal response that implicitly separates defect types. On the synthetic benchmark the calibrated ensemble attains perfect ranking ($\text{AUROC} = 1.000$, $F_1 = 1.000$) with a wide decision margin. Measured on an NVIDIA RTX 4050 laptop GPU, the full ensemble sustains 26.8 FPS (37.3 ms/frame, fp16) in 641 MB of memory—real-time and full-accuracy simultaneously—while an optional single-detector mode reaches 110 FPS for edge use. Finally, we extend the framework from per-frame scoring to batch-level video inspection: a recorded fabric batch is processed frame-by-frame, defects are localized along the batch length (in metres and segments), a minimum-defect-size filter suppresses spurious sub-threshold texture, and a per-batch quality-control record is generated automatically.

The modular architecture, mathematically rigorous formulation, extensive visualization capabilities, and production-grade web dashboard make UltraFabric-Vision suitable for both research investigation and industrial deployment. By open-sourcing the complete framework, we aim to accelerate the adoption of advanced AI-powered quality control in textile manufacturing, particularly for small and medium enterprises that lack resources for proprietary commercial solutions.

The three complementary detection pathways provide robust anomaly detection across diverse defect types through their distinct inductive biases: PatchCore excels at local texture anomaly localization, DINO-v2 captures global structural context through multi-head attention, and ViT-AE detects reconstruction failures from learned normal manifolds. The

ensemble fusion effectively combines these complementary strengths while suppressing individual weaknesses.

Future work will focus on industrial dataset validation, on-line memory bank adaptation, temporal consistency modeling, and edge device deployment through model distillation. The complete open-source implementation is publicly available to foster further research and industrial adoption.

APPENDIX A

DERIVATION OF ENSEMBLE FUSION WEIGHTS

The optimal ensemble weights can be derived through Bayesian model averaging. Given the posterior probability of defect $y = 1$ given input $\mathbf{x}$:

$$p(y = 1|\mathbf{x}) = \sum_{m \in \mathcal{M}} p(y = 1|\mathbf{x}, m)p(m|\mathcal{D}) \quad (38)$$

where $p(m|\mathcal{D})$ is the posterior probability of model m given training data $\mathcal{D}$. Under uniform priors, $p(m|\mathcal{D}) \propto p(\mathcal{D}|m)$, which can be approximated by the model's validation performance. For simplicity and robustness, we adopt uniform weights $w_m = 1/3$, which provides strong baseline performance. Learned weights through Bayesian optimization or neural network meta-learning are left for future work.

APPENDIX B

COMPUTATIONAL COMPLEXITY ANALYSIS

The overall complexity per frame is dominated by the three Transformer forward passes and the two nearest neighbor searches:

- PatchCore ViT-B/16 forward: $\mathcal{O}(L \cdot N \cdot d^2)$ where $L = 12$, $N = 197$, $d = 768$
- PatchCore KNN search: $\mathcal{O}(N_{\text{patch}} \cdot |\mathcal{M}| \cdot d)$
- DINO-v2 ViT-S/8 forward: $\mathcal{O}(L \cdot N \cdot d^2)$ where $L = 12$, $N = 785$, $d = 384$
- ViT-AE forward: $\mathcal{O}(L_{\text{enc}} \cdot N \cdot d^2 + L_{\text{dec}} \cdot N \cdot d^2)$
- Ensemble fusion: $\mathcal{O}(H \cdot W)$ for heatmap resizing and averaging

The GPU-accelerated KNN operations using `torch.cdist` achieve $\mathcal{O}(N \cdot K \cdot d)$ complexity with highly optimized CUDA kernels, providing orders of magnitude speedup over CPU-based alternatives.

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Data Availability Statement: The complete source code, trained model weights, training pipelines, evaluation scripts, and synthetic data generation utilities are publicly available at <https://github.com/varshinicb1/UltraFabric-Vision> under the MIT License.

Conflict of Interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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